

Full Step-by-Step Execution Process (End-to-End)

Step 1: Understand the Objective

This project is not just “plot graphs.”

It answers:

What type of movies perform better?

Which genres dominate?

How do ratings vary over time?

What content trends exist?

This is business intelligence + data-driven decision making.

Step 2: Environment Setup

Install Requirements

```
pip install pandas numpy matplotlib seaborn nltk wordcloud
```

Run Notebook

Jupyter notebook

Open:

IMDb_EDA_Final_Fixed.ipynb

Data Cleaning Process (What You Actually Did)

Real workflow inside notebook:

Load Dataset

```
df = pd.read_csv('imdb_processed.csv')
```

Check Structure

```
df.info()
```

```
df.describe()
```

```
df.isnull().sum()
```

Handle Missing Values

- Remove rows with critical missing data
- Fill numeric columns if required

Remove Duplicates

```
df.drop_duplicates(inplace=True)
```

Text Cleaning (Cast, Genre)

- Remove stopwords
- Tokenization
- WordCloud generation

Charts Used in This Project (And Why)

1. Genre Distribution (Bar Chart)

Chart Type: Bar Plot

Purpose: Count of movies per genre

Real-World Use:

- Production houses decide which genres to invest in
- OTT platforms optimize recommendation categories
- Market demand analysis

2. Ratings Distribution (Histogram)

Chart Type: Histogram

Purpose: Distribution of IMDb ratings

Real-World Use:

- Identify quality spread
- Check if most movies are average or high quality
- Platform credibility analysis

3. Year-wise Trend (Line Chart)

Chart Type: Line Plot

Purpose: Movies released per year

Real-World Use:

- Industry growth tracking
- Detect content boom years
- Pandemic impact analysis

4. Duration Analysis (Boxplot)

Chart Type: Boxplot

Purpose: Distribution of movie lengths

Real-World Use:

- Understand audience attention span
- Compare OTT vs theatre format lengths
- Content format optimization

5. Top Cast WordCloud

Chart Type: WordCloud

Purpose: Most frequent actors

Real-World Use:

- Star power impact
- Casting strategy insights
- Audience preference mapping

6. Rating vs Genre (Grouped Bar / Boxplot)

Purpose: Which genre gets better ratings

Real-World Use:

- Quality benchmarking
- Risk assessment before production
- ROI prediction

What Real-World Problem Does This Project Solve?

This project solves:

Problem 1: Investment Risk

Producers don't know:

- Which genre performs better?
- What rating range attracts audience?

This analysis reduces financial risk.

Problem 2: OTT Recommendation Optimization

Platforms need:

- Popular genres
- High-rated categories
- Trending actors

This improves recommendation algorithms.

Problem 3: Market Trend Analysis

Helps:

- Film analysts
- Streaming platforms
- Media research firms

Understand:

- Industry growth
- Content trends
- Viewer quality expectations

If You Present This in Viva / Interview

You should say:

"This project performs exploratory data analysis on IMDb dataset to identify content trends, rating distribution, genre dominance, and casting influence. It helps production houses and OTT platforms make data-driven decisions regarding content investment and audience targeting."

Power BI Dashboard Structure (Professional Layout)

Page 1: Overview

- Total Movies
- Avg Rating
- Total Genres
- Year Range

Page 2: Genre Analysis

- Genre count
- Avg rating per genre

Page 3: Trend Analysis

- Year-wise release trend
- Rating trend over time

Page 4: Cast Insights

- WordCloud
- Top actors by frequency

How to Improve This Project (Advanced Level)

If you want industry-level improvement:

- Correlation heatmap
- Rating prediction model (ML)
- ROI estimation (if budget data available)

- Clustering of movies
- Sentiment analysis on review